

Piyush Sharma

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Experience

Software Engineer Intern | Synexoo

Jan 2025 – Mar 2025

- Architected and deployed scalable RESTful APIs for a social media platform using Node.js, Express, and MongoDB; implemented JWT-based authentication, role-based access control, and optimized database queries to support growing user traffic.
- Designed follow/unfollow system and profile management modules with efficient schema modeling, contributing to a 30% increase in active user engagement during testing phase.
- Engineered real-time messaging infrastructure using Socket.io (WebSockets), enabling reliable bidirectional communication for 500+ concurrent users with stable session handling and event-driven architecture.

Projects

SentinelAI | NestJS, Redis, Docker

- Formulated a production-grade API security gateway in NestJS that evaluates 90% of incoming traffic using adaptive risk scoring (0–100), dynamically classifying requests as ALLOW, THROTTLE, or BLOCK — eliminating reliance on static rate limiting.
- Configured Redis-based sliding window rate limiting using sorted sets + TTL, auto-blocking malicious IPs after 8–10 abnormal requests; reduced simulated abuse traffic by 92% during attack testing while maintaining zero downtime.
- Modeled a secure admin control plane (X-ADMIN-API-KEY protected) with real-time ban inspection, risk analysis, and manual override APIs; published containerized system via Docker on Railway with production-ready environment isolation.

VCMtalker AI Assistant | Python, Ollama, Gemma, PyInstaller

- Put into production a fully offline desktop AI assistant for Windows (downloadable installer) leveraging a locally hosted LLM (Ollama + Gemma), enabling natural-language voice/text interaction, system automation, and file intelligence without any cloud dependency — ensuring 100% data privacy and sub-second response latency on consumer hardware.
- Built an end-to-end voice interface pipeline (speech-to-text → command engine → local LLM → text-to-speech) supporting always-listening mode, application launching, web automation, and real-time system monitoring, achieving >80% command execution accuracy across 50+ system actions.
- Delivered a production-ready desktop product with persistent long-term memory and OS-level integration, packaged via PyInstaller + Inno Setup into a one-click installer (95MB), enabling users to launch a fully functional AI agent locally in <2 minutes with zero external services.

Graph AI Query System | React, Node.js, SQLite, LLM (Groq), Render

- Engineered and deployed an AI-powered graph query system converting relational ERP data into interactive visual networks, enabling natural language → SQL translation using LLMs and reducing manual query effort by 80%.
- Created a full-stack architecture (React + Node.js + SQLite + REST APIs) integrating real-time graph visualization (force-directed) with optimized query pipelines, improving data interpretability and reducing analysis time by 60%.
- Operationalized intelligent fallback mechanisms and query sanitization, increasing LLM query accuracy from 65% to 90%, while ensuring stable responses across dynamic datasets in production deployment.

FinBoard | Next.js, TypeScript, Recharts, Vercel

- Developed a modular, widget-based stock analytics dashboard using Next.js 16, React 19, and TypeScript, enabling users to build customizable workspaces with drag-and-drop persistence and multi-interval charting.
- Architected a multi-provider API fallback layer (Alpha Vantage + Finnhub + TwelveData) with centralized data normalization and client-side caching, reducing failed API responses by 70% under rate-limit conditions.
- Implemented interactive financial visualizations (line & OHLC candle charts) using Recharts with Zustand-based global state management, ensuring consistent real-time rendering across 3+ data sources; deployed securely on Vercel.

Skills

Languages: C++, Python, JavaScript, TypeScript, SQL.

Backend & Systems: Node.js, Express.js, REST APIs, Redis, MongoDB, MySQL, WebSockets, JWT.

Frontend & CS Fundamentals: React, Next.js, Angular, Object-Oriented Programming, DSA, System Design.

Deploy: Render, Railway, Vercel, Netlify.

DevOps: Docker, AWS, Git, Linux.

AI&Data: Machine Learning (ARIMA, LSTM, Linear Regression), NLP (NLTK VADER), Local LLM Integration (Ollama), LangChain.

Competitive Programming

- **LeetCode** - Rating 1650+ (in top 17.3%), Solved 550+ Questions
- **(3-star) CodeChef** - Rating 1600+, Solved 200+ Questions

Education

Thapar Institute of Engineering and Technology, BTech in Computer Science

Oct 2022 – Jun 2026